SMART CITY BRAKE SUPPORT (SCBS)

id041500200000

Purpose

- The Smart City Brake Support (SCBS) operates the automatic brakes (brake control (SCBS brake)) when the vehicle approaches a vehicle ahead at a vehicle speed between **approx. 4 km/h {2 mph}** to **30 km/h {19 mph}** to reduce collision impact.

Caution

- **Always use tires for all wheels that are of the specified size, and the same manufacture, brand, and tread pattern. In addition, do not use tires with significantly different wear patterns on the same vehicle. If such improper tires are used, the Smart City Brake Support (SCBS) system may not operate normally.**

Note

- The Smart City Brake Support (SCBS) is a system for assisting driver operations. Accordingly, if the conditions for the Smart City Brake Support (SCBS) are met, or even when the Smart City Brake Support (SCBS) is operating, if the driver operates the steering wheel, accelerator pedal, and the brake pedal, the driver's operations take precedence and the Smart City Brake Support (SCBS) operation is canceled.
- Turn off the Smart City Brake Support (SCBS) system to prevent mistaken operation when the vehicle is running on a chassis roller.

Function

- The Smart City Brake Support (SCBS) functions are categorized as follows:
  - — Smart City Brake Support (SCBS) warning alarm which the alarm sounds when the laser sensor detects the vehicle approaching a vehicle ahead or an obstruction. the vehicle is approaching a vehicle ahead.
  - — Brake prefill which reduces the clearance between the brake pads and the disc plate when the laser sensor detects the vehicle approaching a vehicle ahead or an obstruction to enable brake force generation immediately when the brake is operated.
  - — Brake control (SCBS brake) which operates the automatic brakes when the laser sensor detects the risk of a collision.
- When the laser sensor detects that the vehicle is approaching a vehicle ahead or an obstruction, the system operates the brake prefill and sounds the warning alarm. When the driver depresses the brake pedal, the brakes are applied firmly and quickly to assist, and if the driver does not perform an avoidance maneuver, it operates the brake control (SCBS brake).

Warning

- **The Smart City Brake Support (SCBS) performs the brake control (SCBS brake), however, it is not a system which guarantees collision prevention under all conditions. Because deceleration by the brake control (SCBS brake) is limited, always verify the safety of the surrounding area and keep a safe distance from a vehicle ahead.**
- **The Smart City Brake Support (SCBS) does not operate when the vehicle speed is less than approx. 4 km {2 mph} or approx. 30 km/h {19 mph} or more. It also does not operate when the vehicle is in reverse.**
- **Do not apply stickers to the surface of the windshield near the laser sensor (including transparent stickers). Otherwise, the laser sensor may not be able to detect vehicles or obstructions ahead which could result in an accident.**

Smart City Brake Support (SCBS) display

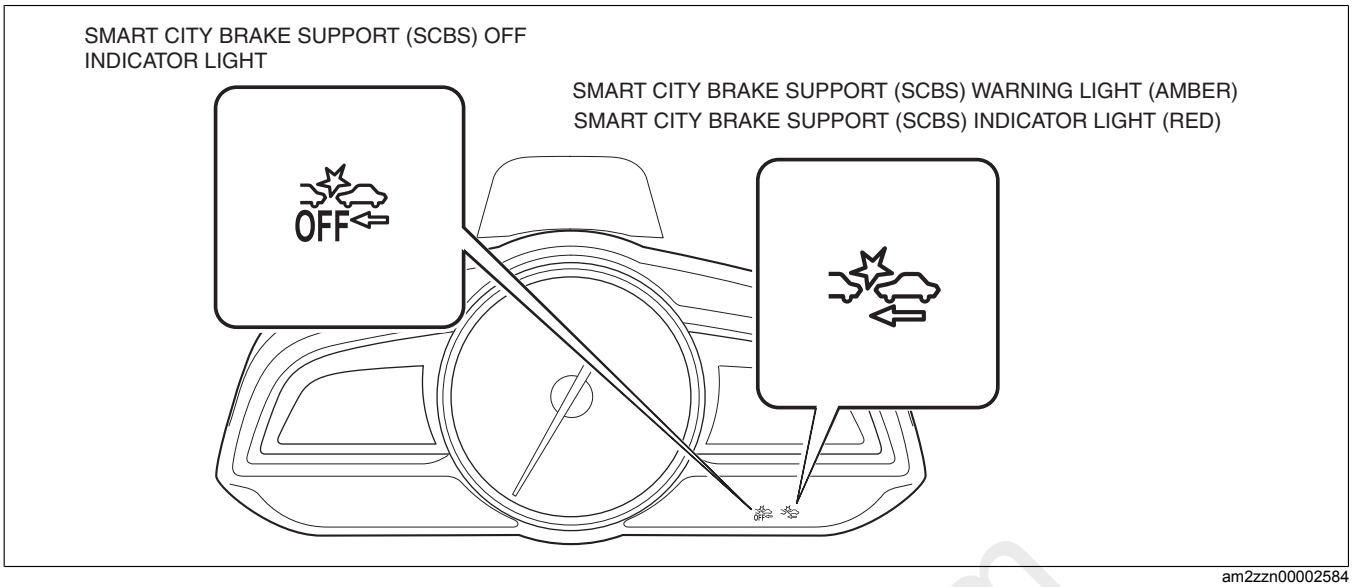
- The Smart City Brake Support (SCBS) display which displays in the active drive display or the multi-information display is as follows:

Note

- For vehicle with center display, a warning message in the center display is displayed with the Smart City Brake Support (SCBS) warning light (amber) turned on or flashed. For the message content and verification method, refer to the CENTER DISPLAY. (See CENTER DISPLAY [WITH CENTER DISPLAY].)
- When the Smart City Brake Support (SCBS) system operates, the user is notified by the flashing warning light and the warning message.

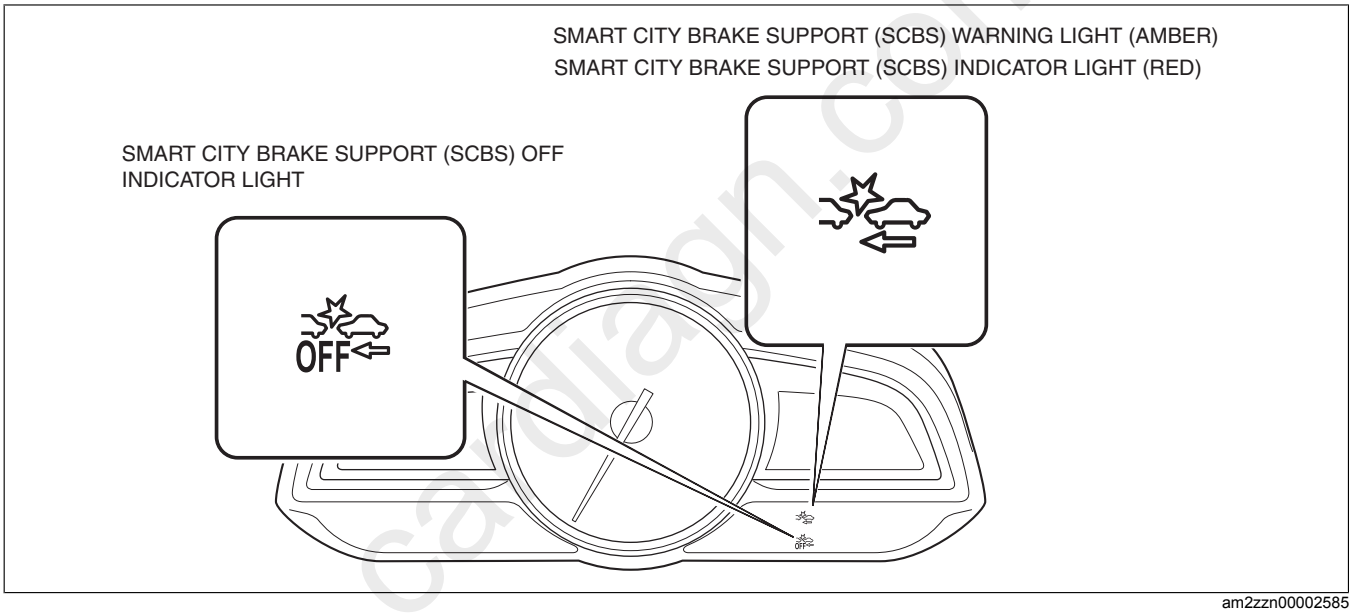
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Digital speedometer type



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Analog speedometer type



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Smart City Brake Support (SCBS) operating condition		Content	Smart City Brake Support (SCBS) indicator light (red)	Smart City Brake Support (SCBS) warning light (amber)	Smart City Brake Support (SCBS) OFF indicator light	Active driving display*
Disable	After switch the ignition ON (engine off or on)	Smart City Brake Support (SCBS) is disabled	Not illuminated/Not flashed	Not illuminated/Not flashed	Illuminated	No display

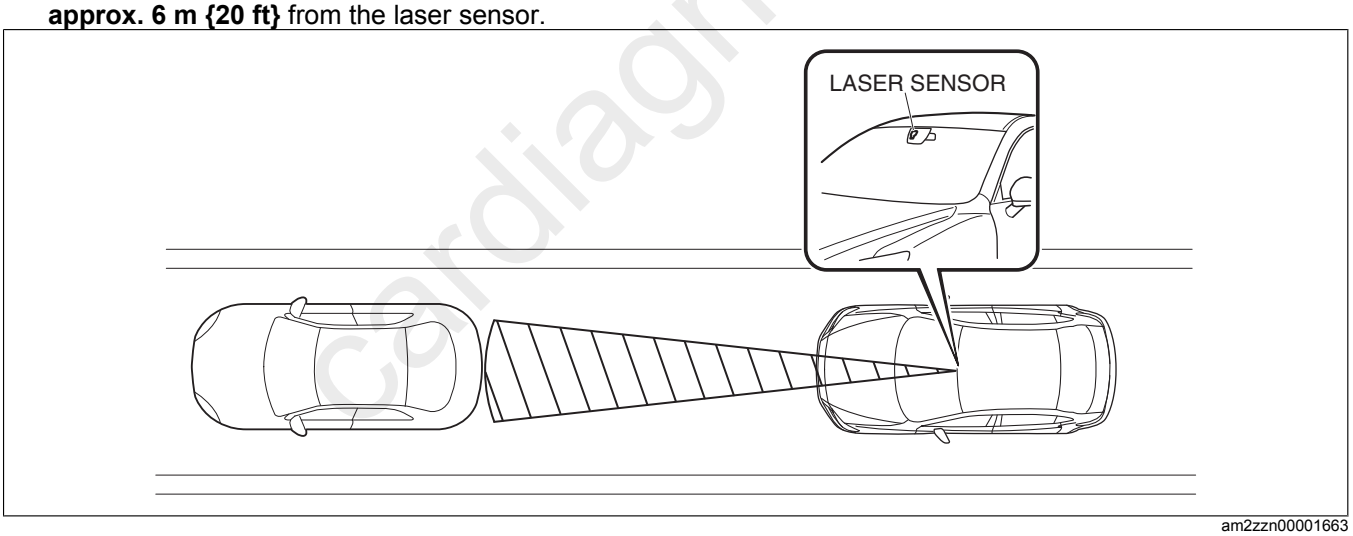
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Smart City Brake Support (SCBS) operating condition		Content	Smart City Brake Support (SCBS) indicator light (red)	Smart City Brake Support (SCBS) warning light (amber)	Smart City Brake Support (SCBS) OFF indicator light	Active driving display*
Enable	When there is the possibility of a collision with a vehicle ahead or an obstruction	Brake control (SCBS brake) is in operation	Flashes	Not illuminated/Not flashed	Not illuminated/Not flashed	SCBS Automatic Brake
	When a malfunction occurred in the Smart City Brake Support (SCBS) system	The Smart City Brake Support (SCBS) system malfunction is occurred	Not illuminated/Not flashed	Illuminated	Not illuminated/Not flashed	No display
		The laser sensor capability is impaired by rainfall, snowfall, dirty windshield and so forth.	Not illuminated/Not flashed	Illuminated	Not illuminated/Not flashed	No display

* : If equipped.

Construction/Operation

- The laser sensor installed on the upper area of the windshield detects a vehicle ahead or an obstruction within **approx. 6 m {20 ft}** from the laser sensor.



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- The Smart City Brake Support (SCBS) system will operate under the following conditions:
  - — The engine is running.
  - — The center display in the instrument cluster does not indicate that the system requires inspection.
  - — The vehicle speed is between **approx. 4 km/h {2 mph}** to **30 km/h {19 mph}**.
  - — The Smart City Brake Support (SCBS) system is not turned off.
  - — The DSC is not malfunctioning.
  - — The driver is not deliberately performing driving operations (accelerator pedal, steering wheel, and braking operations).

Note

- The smart city brake support (SCBS) system will operate even if the DSC OFF indicator light is illuminated after the DSC is stopped by operating the DSC OFF switch. If the smart city brake support (SCBS) system operates while the DSC has been stopped, the DSC becomes operational and the DSC OFF indicator light turns off.
- The Smart City Brake Support (SCBS) detects a vehicle ahead by emitting a near infrared laser beam and receiving the beam reflected off the reflector or the vehicle ahead, and then using it for the measurement. Consequently, the Smart City Brake Support (SCBS) may not operate under the following conditions:
  - — Reflection of laser sensor is poor due to the shape of the vehicle ahead.

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— The vehicle ahead is significantly dirty.
— Under bad weather condition such as rain, fog, and snow.
— When the window washer is being used or the windshield wipers are not used when it is raining.
— The windshield is dirty.
— The steering wheel is turned completely left or right, or the vehicle is accelerated rapidly and comes close to the vehicle ahead.
— Trucks with low loading platforms, vehicles traveling at extremely low speeds, and vehicles with high profile.
— Vehicles with certain shapes such as a vehicle carrier.

- Under the following conditions, the Smart City Brake Support (SCBS) system may not operate normally:
  — Heavy luggage is loaded in the luggage compartment or on the rear seat.
  — When there is the possibility of a partial contact with a vehicle or obstruction ahead.
  — When driving on continuously curving roads, entering and existing curves, and unstable driving due to a vehicle accident or breakdown in a driving lane.
  — Elongated luggage or cargo is loaded onto installed roof rails and covers the laser sensor.
  — Exhaust gas from the vehicle ahead, sand, snow, and water vapor rising from manholes and grating, and water splashed into the air.
- In the following cases, if the laser sensor determines that there is a vehicle ahead and the Smart City Brake Support (SCBS) system may operate:
  — Object on the road at the entrance to a curve.
  — Vehicles passing in the opposite lane while making a curve.
  — Metal objects, bumps, or protruding objects on the road.
  — When passing through a toll gate.
  — When passing under a vinyl curtain or flag.
  — Plastic objects such as pylons.
  — Two-wheeled vehicles, pedestrians, animals or standing trees.
  — There is fogging on the inner surface of the windshield (where the near infrared laser beam is emitted).
    - Operate the air conditioning system to defrost the windshield around the laser sensor first before starting to drive.
  — There is/are fogging/water drops on the outer surface of the windshield (where the near infrared laser beam is emitted).
    - Operate the wiper system to remove the fogging/water drops on the windshield.

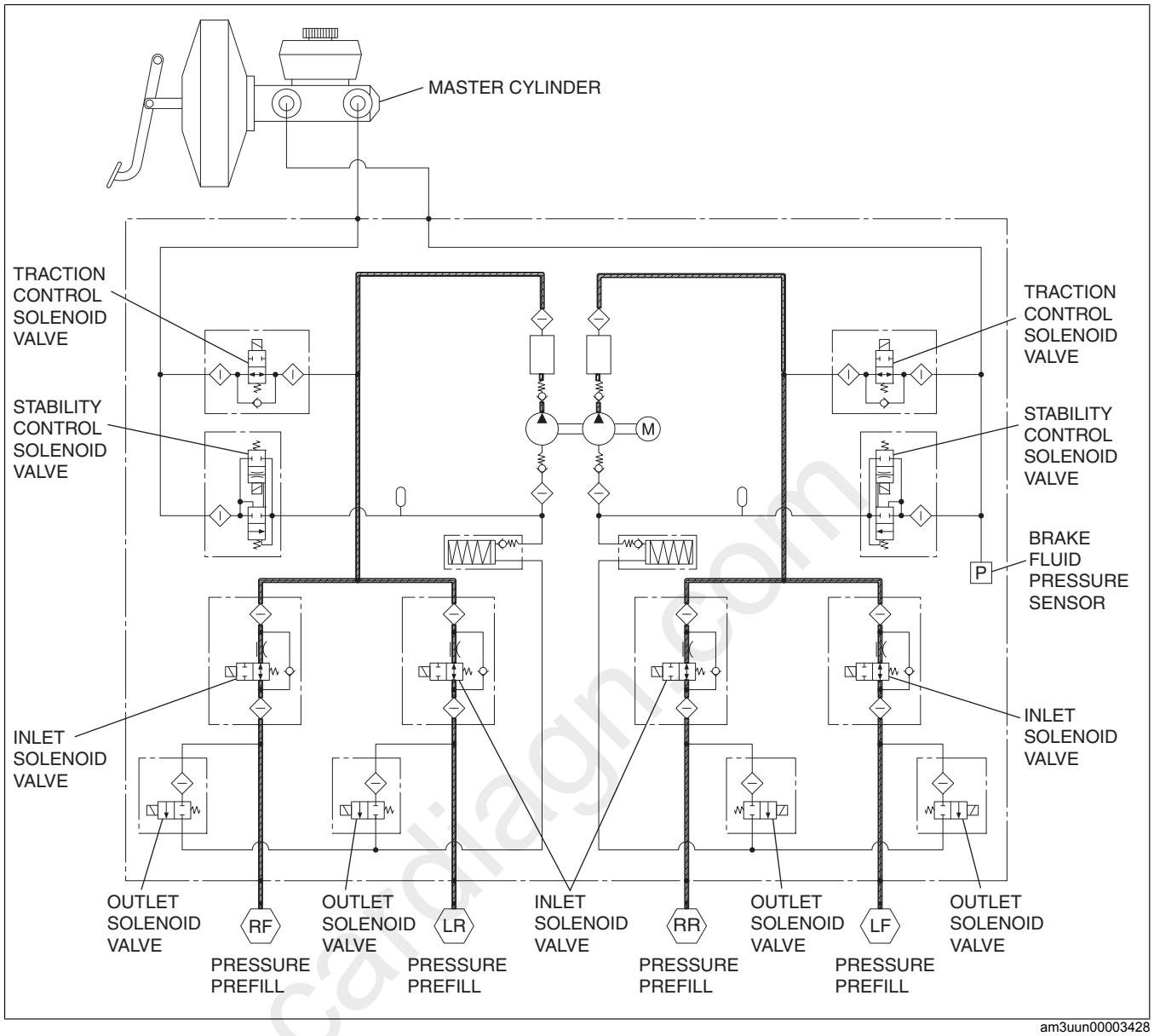
Brake prefill operation

Solenoid valve operation table

Traction control solenoid valve		Stability control solenoid valve		Inlet solenoid valve				Outlet solenoid valve				Pump motor, pump
LF—RR	RF—LR	LF—RR	RF—LR	LF	RF	LR	RR	LF	RF	LR	RR	
OFF (open)		OFF (close)		OFF (open)				OFF (close)				Operating

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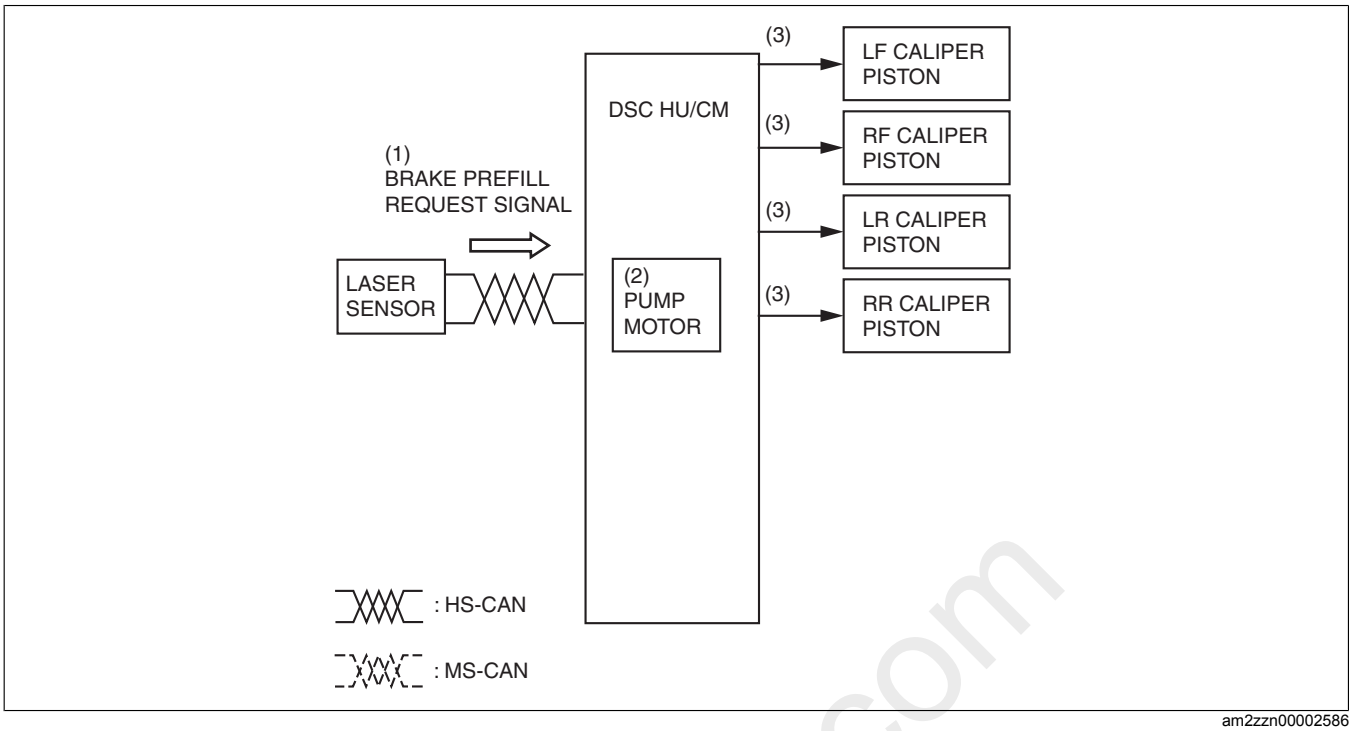
Hydraulic circuit diagram



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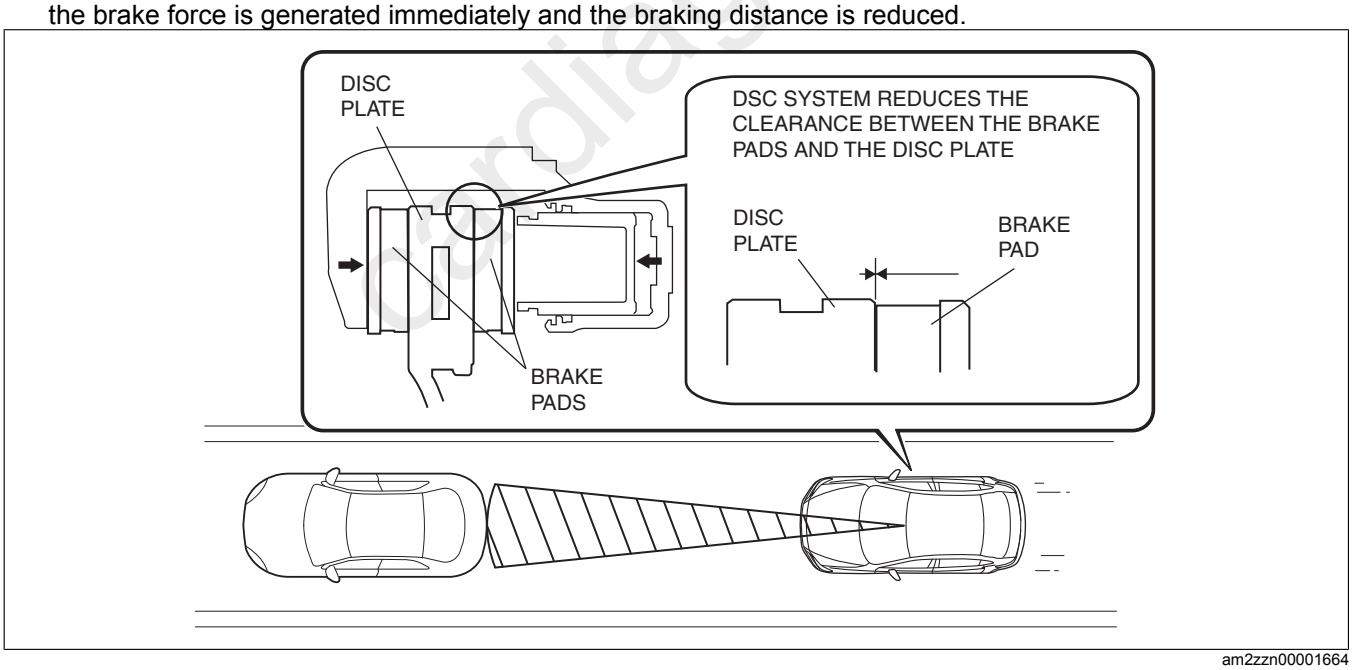
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Block diagram



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- When the laser sensor detects that the vehicle is approaching a vehicle ahead or an obstruction, it outputs a brake prefill request signal (1) to the DSC HU/CM.
- The DSC HU/CM operates the pump motor (2) to activate the pump so that the brake fluid pressure is lead to the caliper piston (3) and the brake pads are kept near the disc plate.
- As a result, if the driver realizes that he/she is approaching a vehicle ahead or an obstruction, and depresses the brake pedal, or if the driver does not perform an avoidance maneuver and the automatic brakes are operated, the brake force is generated immediately and the braking distance is reduced.



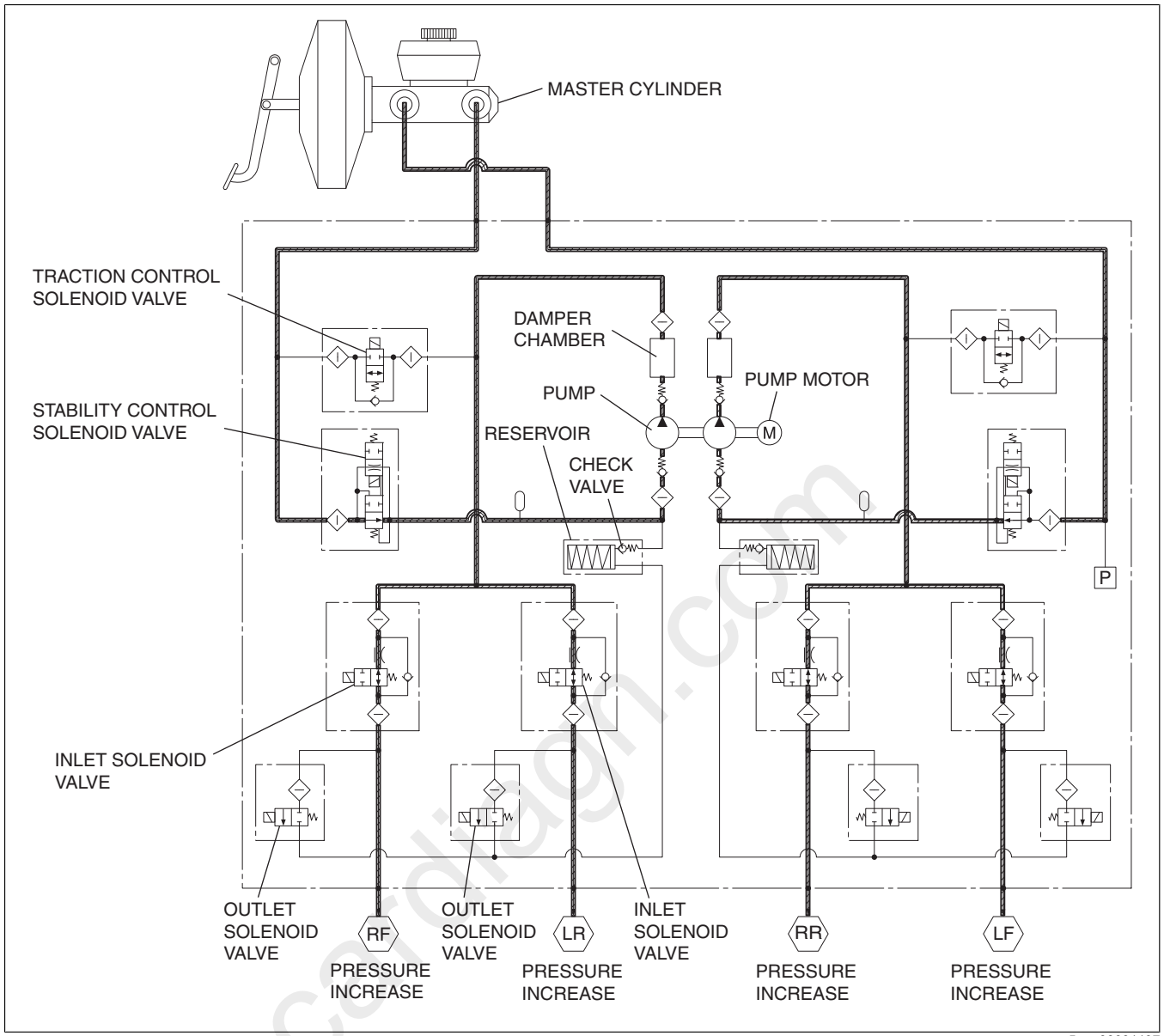
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Brake control (SCBS brake) operation

Solenoid valve operation table

Traction control solenoid valve		Stability control solenoid valve		Inlet solenoid valve				Outlet solenoid valve				Pump motor, pump
LF—RR	RF—LR	LF—RR	RF—LR	LF	RF	LR	RR	LF	RF	LR	RR	
ON (close)		ON (open)		OFF (open)				OFF (close)				Operating

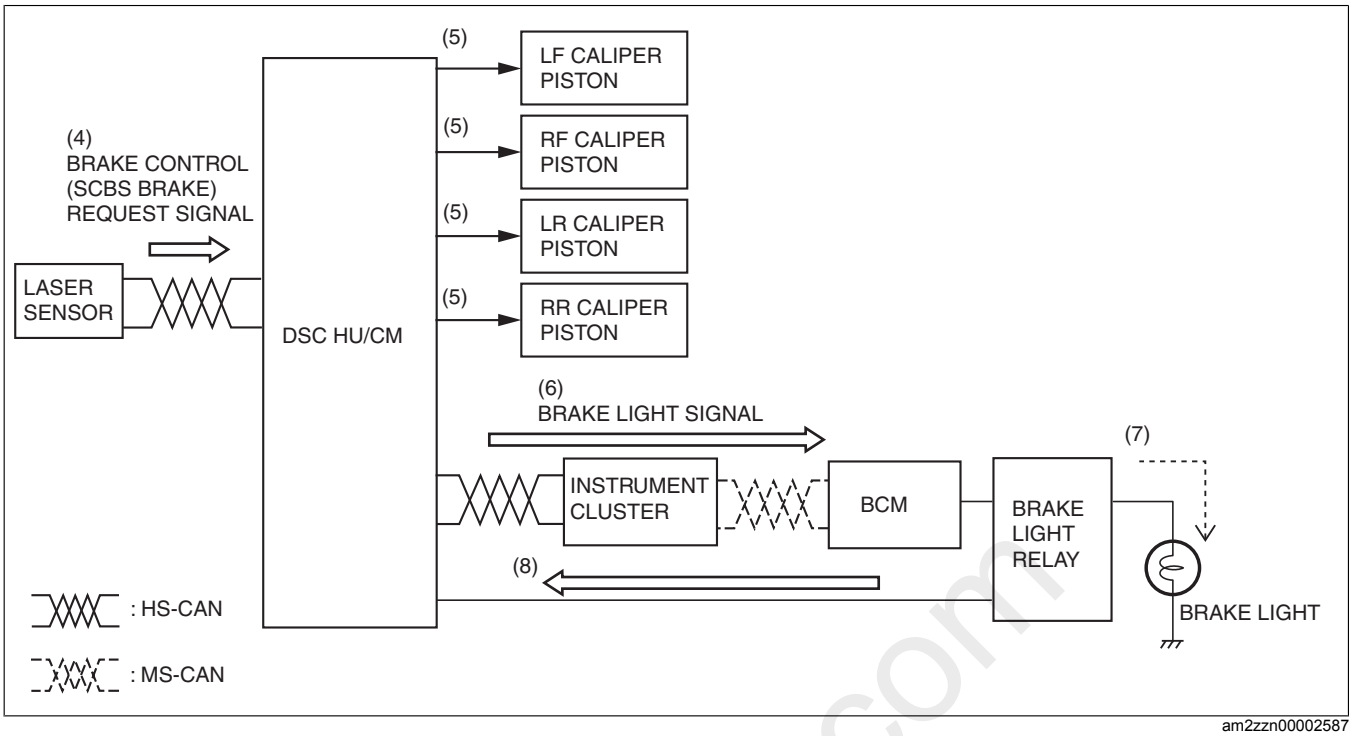
Hydraulic circuit



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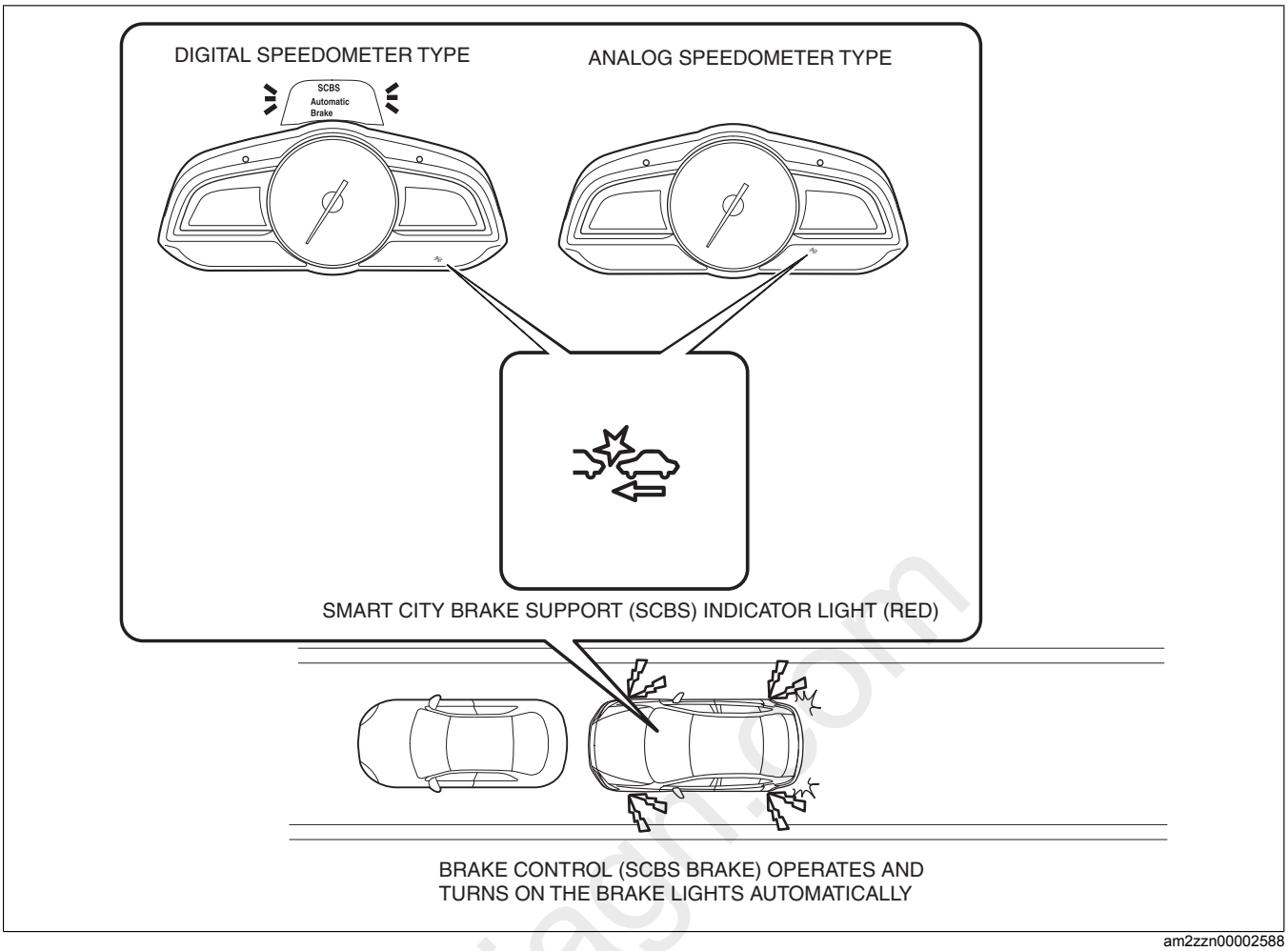
Block diagram



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- After the brake prefill operates, if the driver does not perform an avoidance maneuver and the laser sensor determines that a collision may occur, it outputs a brake control (SCBS brake) request signal (4) to the DSC HU/CM.
- The DSC HU/CM energizes the traction control and the stability control solenoid valves to switch the hydraulic circuits and to increase the brake fluid pressure. Therefore the brake is operated automatically (5). At this time, the DSC HU/CM outputs a brake light signal to the Body Control Module (BCM) (6). While the brake control (SCBS brake) is operating, turns on the brake lights automatically (7), and at the same time the brake light unit outputs a brake light operation signal to the DSC HU/CM (8).
- When the Smart City Brake Support (SCBS) operates and the brakes are applied, the Smart City Brake Support (SCBS) indicator light (red) is flashed and the message “SCBS Automatic Brake” is indicated in the active drive display (instrument cluster type A) or multi-information display (instrument cluster type B).
- The brake control (SCBS brake) is kept operated for **2 s.** To keep the vehicle stopped after the brake control (SCBS brake) operation ends, the brake pedal has to be depressed to operate the brake.

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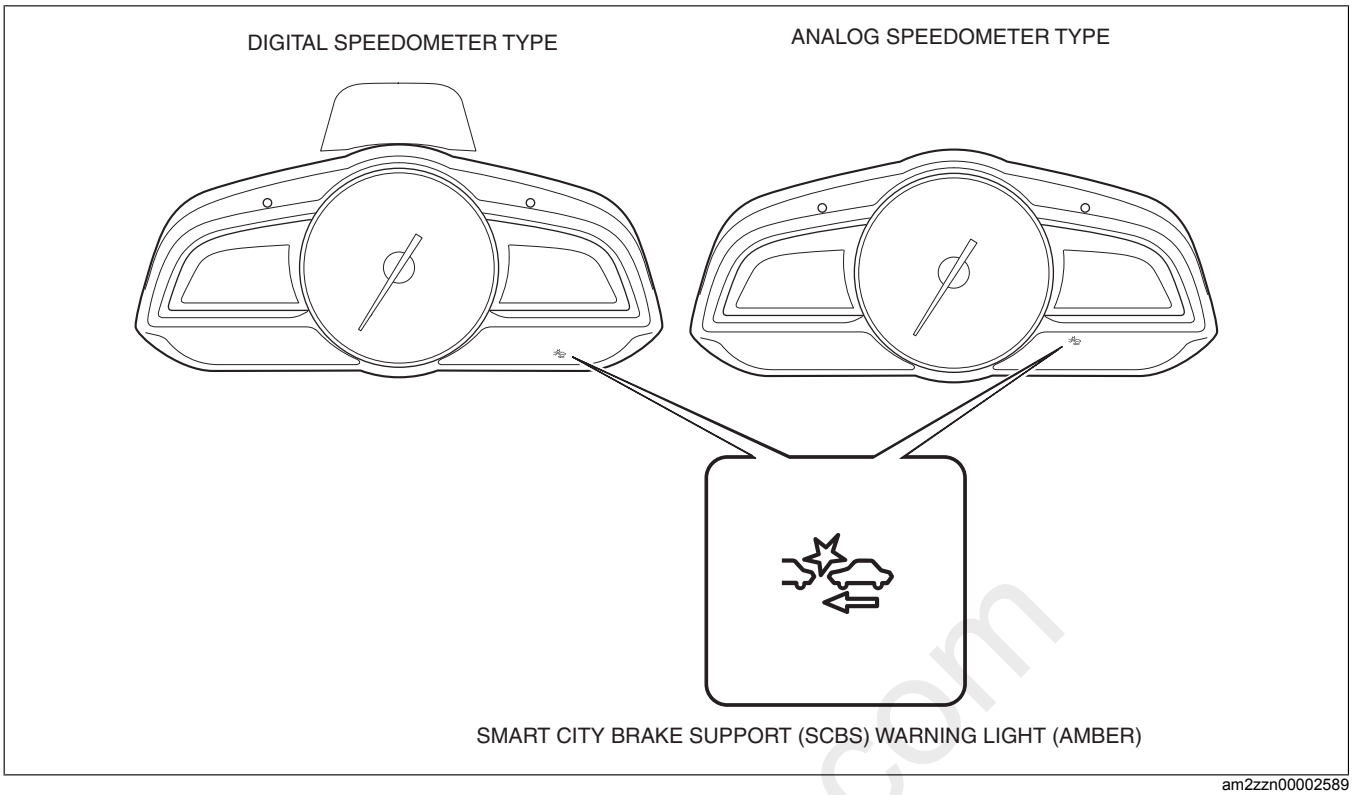
Caution

- **When the laser sensor malfunction is determined as a result of the on-board diagnostic test, the Smart City Brake Support (SCBS) is suspended while driving.**
- **When the DTC is stored in the PCM, DSC HU/CM, and/or instrument cluster, the Smart City Brake Support (SCBS) is suspended depending on the malfunctioning situation.**

Note

- If the Smart City Brake Support (SCBS) system has a malfunction and the Smart City Brake Support (SCBS) control is inhibited, the Smart City Brake Support (SCBS) warning light (amber) illuminates to alert the driver.

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